



Common Phase 1

Submission and Evaluation Guidelines

AESS Sustainability Hackathon 2026



Important note.

This file collects the shared Phase 1 rules that apply across all four challenges, including deliverables, repository expectations, naming rules, review workflow, and integrity requirements. Challenge-specific technical details remain in the individual challenge files.

1. Phase 1 Overview

- Phase 1 is the online hackathon stage in which participating teams develop technical solutions addressing one of the four sustainability challenges.
- The objective of this phase is to evaluate technical capability, innovation, implementation quality, and sustainability integration in order to identify the top teams that will advance to the onsite finals.
- Teams work remotely during the preparation and development period, then submit a complete project package for technical review.

2. Phase 1 Timeline

- Phase 1 Launch: 01 May 2026
- Submission Deadline: 15 May 2026
- Technical Review Period: 15–21 May 2026
- Results Announcement: 22 May 2026

3. Team Eligibility

- Teams must consist of 3–5 participants.
- Eligible participants include undergraduate students, graduate students, and early-career engineers.
- Participation is open globally.
- IEEE membership is not mandatory for Phase 1, but it may be required for finalists seeking travel support for Phase 2.

4. Mandatory Deliverables for Every Challenge

- Project Repository (GitHub or equivalent, public or jury-accessible).
- Concept Document (maximum 2 pages, PDF).
- Demo Video (2–3 minutes).
- Presentation Deck (PDF).
- Supporting Code / Simulation / Schematics / Technical Files as applicable.

5. Submission and File-Naming Rules

- Submissions must be made through the official submission form, Google Form, or official event portal used by the organizers.
- Repository links, direct uploads, and cloud-storage links may be used according to the submission system.
- All submitted files must follow the naming convention: TeamName_ChallengeTitle_Phase1_FileType.
- Examples include: EcoSat_SpaceElectronics_Phase1_Presentation.pdf, EcoSat_SpaceElectronics_Phase1_Code.zip, and EcoSat_SpaceElectronics_Phase1_Video.mp4.

6. Shared Repository Expectations

- The repository must contain all code, simulations, documentation, and supporting files required to reproduce and evaluate the project.
- Teams should organize the repository clearly, using a structure such as /README.md, /docs/, /src/, /simulation/, /results/, /assets/ (optional), and /hardware/ (if applicable).
- The README should explain the project overview, problem statement, sustainability strategy, how to run the simulations, required tools and dependencies, and the challenge number.
- Code quality matters: the technical jury will look for readable organization, meaningful naming, comments where needed, modular structure, and reproducibility of results.

7. AI Usage and Reproducibility

- AI tools may be used for code assistance, optimization suggestions, documentation drafting, or simulation guidance.
- All AI-assisted outputs must be validated, understood, and defensible by the team.
- Teams should clearly disclose AI usage in the repository README when it has meaningfully contributed to the work.
- Judges must be able to reproduce key results, run main simulations, verify major calculations, and understand the assumptions used.

8. Evaluation Process

- Each complete submission is screened first for completeness and submission compliance.
- Compliant submissions are then evaluated by a technical jury panel that may include aerospace engineers, radar and sonar specialists, and sustainability experts.
- Each project is typically reviewed by 2-3 judges.
- Final scores are averaged to determine rankings and finalist selection.

9. Common 100-Point Scoring Framework

- Technical Innovation – 25 points
- Technical Feasibility – 20 points
- Sustainability Impact – 20 points
- Implementation Evidence – 15 points
- Documentation Quality – 10 points
- Presentation Quality – 10 points

10. Academic Integrity and Disqualification Rules

- All submitted work must be original.
- External sources must be cited when used.
- Teams may be disqualified for plagiarism, missing mandatory deliverables, invalid or misleading results, or unverifiable AI-generated content.
- A demonstrated inability to explain the technical content of the submission may negatively affect evaluation and can become grounds for disqualification in serious cases.

11. Finalist Selection and Communication

- The top-ranked teams will be selected as finalists for the onsite phase, with approximately ten finalist teams expected, subject to organizer discretion and submission quality.
- Participants receive official updates through event email, the event website, and official social media announcements.
- Selected finalist teams will be invited to participate in the onsite finals event in Egypt.

12. Quick Submission Checklist

Item	Item
Repository link accessible	☐
Concept document attached	☐
Demo video attached or linked	☐
Presentation deck attached	☐
Supporting technical files attached	☐
Correct file naming used	☐
README explains tools and dependencies	☐